

RAHUL MITRA

+65 8515 5413 | mitrasahul2002@gmail.com | linkedin.com/in/rahulmitra-dev | rahul-mitra.com | github.com/Rah-Rah-Mitra

EDUCATION

National University of Singapore

Aug 2023 – Jul 2027

Bachelor of Engineering, Industrial Systems Engineering

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.
- Relevant coursework: Software Engineering, Data Structures and Algorithms, Artificial Intelligence, 3D Computer Vision, Operations Research, Simulation, Statistics, Linear Algebra.

WORK EXPERIENCE

Operations Research (NUS System Design Project), Contract, STMicroelectronics

Aug 2026 – Present

- Design an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, replacing experience-based slotting with capacity-aware decisions drawn from 3 live inputs: picking history, demand forecasts, and available capacity.

Robotics Vision Engineer, Amazon (Blended AI+X)

Jul 2026 – Present

- Build vision systems for 3 imaging workstreams: camera ISP enhancement, super-resolution, and image restoration under motion and low light.

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories

Jun 2026 – Present

- Assemble a 5-stage offline process-intelligence platform for globally distributed spray-drying plants: GPU OCR and layout parsing, recursive chunking, ONNX embeddings, a Chroma vector store, and LangChain over a locally hosted Gemma model.

Platform & Solutions Engineer (Sparks Citizen Developer), People's Association

Sep 2025 – Sep 2026

- Delivered a digital garden-plot balloting platform across admin and public frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC) to secure resident sign-in and Myinfo to verify identity and address, keeping garden-plot balloting fair for 150 residents signed up at release.
- Engineered keyless GitHub Actions CI/CD on IAM OIDC federation across 5 pinned service repos, with gated Fargate migrations, mandatory RDS snapshots, and circuit-breaker auto-rollback.
- Provisioned AWS infrastructure with Terraform across 8 services: ECS Fargate, Multi-AZ RDS Postgres with PostGIS, ALB fronted by WAFv2, Route 53, ACM, KMS, and Secrets Manager.

Data Analytics & Product Management Intern, Abbott Laboratories

Jan 2026 – Jun 2026

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, then compared 3 approaches (heuristic, MIP, and genetic algorithm) to turn production objectives and constraints into decision-ready schedules.
- Owned cloud deployment and production support for a team-built APC simulator used in manufacturing and engineer training, with Docker packaging, Azure App Service hosting, and documented handover.

PROJECTS

Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams)

Jan 2026

- Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once, placing top 8 of 50+ teams.

LEADERSHIP AND ACTIVITIES

NTUC Health Active Ageing Centre, Elderly Care Volunteer

Sep 2025 – Sep 2026

- Coordinate weekly senior-engagement activities, building empathy for healthcare, accessibility, and community-focused product design.

SKILLS AND CERTIFICATIONS

Technical: Solution architecture, AWS (ECS Fargate, RDS, ALB, WAFv2, Route 53, ACM, KMS, Secrets Manager), Terraform, AWS SAM, Azure, Docker, Bitbucket CI/CD, API design, Singpass Login & Myinfo (OIDC), RAG architecture, vector search, AI tools (Claude Code, ChatGPT, Codex, Gemini, ComfyUI), Supabase, GIS mapping, stakeholder discovery, manufacturing systems

Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery; Deep Reinforcement Learning Hands-On (Packt)